

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for the correct answer.</i> • $(2.9 - 0.7) / 2.9 \times 100 = 75.9\%$ (accept 76%) (1)	1
01.2	<i>1 mark for each point.</i> • Although the larger vehicle uses more energy in total, it carries many more people / a larger load (1) • So the energy per passenger (or per tonne of goods) is lower; steel wheels on rails also have low rolling resistance (1)	2
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Flexibility of routes / destinations and door-to-door convenience (1) • Time / speed and reliability where public transport is infrequent (1) • Low fuel tax / cheap fuel (1) • Carrying goods; comfort; status (1)	3
01.4	<i>1 mark for each point.</i> • Integrated systems (park and ride, bicycles on trains, linked timetables) make it convenient to use the most efficient mode for each part of a journey (1) • So more journeys shift from cars to bulk transport / cycling (1) • ATM / smart motorways use variable speed limits to prevent congestion and keep traffic flowing freely (1) • Vehicles avoid stop-start driving and run nearer their optimum speed (about 56 mph), reducing fuel wasted in acceleration and idling (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-228; AQA spec 3.3.4.2.1.	

Question	Marking guidance	Marks
02.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Asbestos: fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma; disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) (2) - Cyanide: very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested; disposal: incineration so carbon and nitrogen are separated and oxidised to CO₂ and NO_x (much smaller problem) (2) - High-level radioactive waste: intense ionising radiation, heat generation, long half-lives; disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores (2)	6
02.2	<i>1 mark.</i> The fibres are bound / cannot become airborne so cannot be inhaled (1)	1
02.3	<i>1 mark for each point.</i> - Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material (1) - The solid cement immobilises the waste so it cannot leak / leach out (1)	2
02.4	<i>1 mark.</i> Even if the glass shatters the waste stays encapsulated in the fragments; glass is more stable / resistant to heat and leaching over very long timescales (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	
03.1	<i>1 mark for each point.</i> - Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron (1) - A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) (1) - The heat boils water to steam which drives turbines and generators (1)	3
03.2	<i>1 mark for each point.</i> - Output changes slowly (days) so it cannot follow rapid fluctuations in demand (1) - Very high construction cost but low fuel cost means it is most economic running continuously at full output (1)	2
03.3	<i>1 mark for each point.</i> - Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal (1) - Still need condenser cooling water (coast / large lake) and a stable site (1)	2
03.4	<i>1 mark.</i> Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy; reactor construction and decommissioning have embodied energy (1)	1
03.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates (1) - Large projects often overrun original cost estimates (1) - Long-term waste disposal costs uncertain (1) - State subsidies may hide costs (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	
04.1	<i>1 mark.</i> • The time taken for half of the atoms of a radioactive isotope to decay (activity halves) (1)	1
04.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $32 / 8 = 4$ half-lives (1) • $6400 / 2^4 = 400$ Bq (1)	2
04.3	<i>1 mark for each point.</i> • Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient (1) • Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low (1) • Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) (1)	3
04.4	<i>1 mark for each point.</i> • Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure (1) • Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) (1) • Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) (1)	3
04.5	<i>1 mark.</i> • A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
05.1	<i>1 mark.</i> • The time taken for half of a pollutant (released into the environment) to break down / degrade (1)	1
05.2	<i>1 mark for each point.</i> • DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) (1) • DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level (1) • So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level (1)	3
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • The gas is denser than air so it settled close to the ground / did not rise and disperse (1) • Low wind velocity (at night) meant it was not dispersed / diluted (1) • Short half-life so it broke down before travelling far (1)	2

Question	Marking guidance	Marks
05.4	<i>1 mark for each point.</i> • High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) (1) • But dispersal dilutes the pollutant so the concentration at any one point may be lower (1)	2
05.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Noise / sound (1) • Heat / thermal pollution (1) • Ionising radiation (1) • Light (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
06.1	<i>1 mark for each point.</i> • The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter (1) • Current recharge is little or none so any abstraction reduces the stored volume (1) • The large volume means shortages take a long time to appear, but eventually the supply runs out; ancient water is often saline, causing soil salinisation (1)	3
06.2	<i>1 mark for each point.</i> • Two satellites whose orbits are affected by the force of gravity (1) • The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass (1) • Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas (1)	3
06.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Low-pressure irrigation pipes to reduce leaks (1) • Growing crops that need less water than wheat / maize (1) • Charging consumers by volume used (metering) (1) • Inter-basin transfer of water 450 km from the Yangtze (1)	3
06.4	<i>1 mark.</i> • Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	
07.1	<i>1 mark for each point.</i> • The climate naturally fluctuates (solar output, Earth's orbit) so extreme events can occur without human influence (1) • An individual event, or even several, does not show a trend; the change may be in the frequency of events which can only be assessed years later (1) • Changes occur at different time and spatial scales (local storm vs global trend); interconnected systems make attribution complex (1)	3

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> • Towns have a heat-island effect (heat emissions, dark surfaces absorbing light) (1) • As the town grows around the weather station the warming recorded is partly local, not global; older equipment / sparse global coverage also reduce reliability (1)	2
07.3	<i>Up to 5 marks in total; award marks in proportion to the completeness of each point.</i> • Tree rings: width of each annual ring shows the growth rate, indicating temperature / rainfall in that year; rings easily dated by counting; limitation: growth also affected by other factors (water, pests, shading), only covers the life of the tree / limited regions (up to 2) • Ice cores: annual snow layers compact into ice; air bubbles trapped in the ice give the CO₂ (and methane) concentration when trapped; oxygen isotope ratios indicate temperature; up to 800 000 years (Dome C); limitation: deep layers compressed / less distinct, dating needs radio-isotope analysis and volcanic ash layers; only polar regions (up to 2) • Others accepted: coral growth rings; pollen in lake sediments dated by radiocarbon (up to 2)	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 100-102; AQA spec 3.2.1.2.	
08.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $\Sigma n(n - 1) = (40 \times 39) + (30 \times 29) + (20 \times 19) + (10 \times 9) = 1560 + 870 + 380 + 90 = 2900$ (1) • $N(N - 1) = 100 \times 99 = 9900$ (1) • $D = 9900 / 2900 = 3.41$ (accept 3.4) (1)	3
08.2	<i>1 mark for each point.</i> • Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B (1) • The index does not identify which species are present / a rare or important species may be missed; depends on identification skill and sampling method (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
09.1	<i>1 mark for each point.</i> • Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) (1) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites (1)	2
09.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sea current velocity and direction (1) • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate (1) • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people (1) • Geology / hydrology of the coast; sediment deposition areas (mud flats) (1) • Human activities in the area: fishing, seaweed collection, beach use (1)	3

Question	Marking guidance	Marks
09.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) (1) • Monitor their exposure / the concentration of radioactive material in the foods they eat or their body (1) • If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled (1) • CGM assesses exposure before health effects occur (not used to detect damage) (1)	3
09.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Seawater (1) • Sediment / mud (1) • Seaweed (1) • Shellfish / molluscs / fish (1) • Milk, vegetables or meat from nearby farms (1) • Atmospheric dust (1) • Soil (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates; bacterial growths cover the bed, blocking light - Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water; decay of organic matter releases inorganic nutrients - Inorganic nutrients: nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation - Human health: nitrites and methaemoglobinaemia; possible carcinogen via nitrosamines - Control of organic wastes: sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) - Control of nutrients: tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV; Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management; dredging sediments - Evaluation: sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage; tertiary treatment is expensive so only used at sensitive sites; diffuse agricultural nitrate is hard to control and aquifers respond slowly; phosphate legacy in sediments - Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9
10.2	<i>1 mark.</i> Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25

Question	Marking guidance	Marks
11.2	To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society? Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • Renewable features: intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal); low energy density; location; low pollution but manufacturing impacts • Technologies and developments for each resource with examples; new technologies improving usability • Storage and grid management: pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving; the role of electricity as a secondary fuel • Economics and politics: development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution; sustainability principles - natural systems run on low-density renewable energy • Judgement: electricity largely yes; heat, aviation, shipping and high-temperature industry harder; transition period with nuclear / CCS Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	25